

Embedded Systems Programming

College of Science and Technology

School Of ICT

Computer Science Department

Lecturer : NGABO Desire



LECTURE 2



Requirements of Embedded system



1. Reliability
2. Cost-effectiveness
3. Low power consumption
4. Efficient use of processing power
5. Efficient use of memory
6. Appropriate execution time

Reliability



- ◆ Embedded system have to work without the need for resetting or rebooting.
 - ◆ This call for a very reliable hardware and software.
 - ◆ For example : if an embedded system comes to a halt because of hardware error, the system must reset itself without the need for human intervention.
- ! The embedded software developers must make the reliability of the hardware as well as that of the software.**

Cost-effectiveness



- ◆ If an embedded system is designed for a very special purpose such as for deep space or for nuclear power plant station or any utility application
 - ◆ cost may not be an issue.
- ◆ if the embedded system is designed for a mass market purpose like CD players, toys and mobile devices
 - ◆ cost is a major concern.
- ◆ **Application Specific Integrated Circuit (ASIC) is used by the designers to reduce the hardware components and hence the cost also reduces.**

Low power consumption



- ◆ Most of the embedded systems are powered by battery, rather than a main supply.
 - ◆ The power consumption should be minimized to avoid draining the Batteries.
 - ◆ For example :
 - ◆ by reducing the number of hardware component the power consumption can be reduced.
 - ◆ As well as by designing the processor to revert to low power or sleep mode when there is no operation to perform

Efficient use of processing power



- ◆ A wide variety of processors with varying processing powers are available to embedded systems.
- ◆ Developers must keep processing power, memory and cost in mind while choosing the right processor.
- ◆ The processing power requirement is specified in ,Million Instruction Per Second (MIPS).
- ◆ With the availability of so many processor, choosing a processor has become a tough task nowadays

Efficient use of memory



- ◆ Efficient use of memory Most of the embedded systems do not have secondary storage such as hard disk.
- ◆ The memory chip available on the embedded systems are only Read Only memory and Random Access memory.
- ◆ As most of the embedded systems do not have secondary storage, —flash memory is used to store the program.
- ◆ Nowadays micro-controller and Digital signal processors also comes with onboard memory.
- ◆ Such processors are used for small embedded system as the cost generally is low and the execution generally is fast

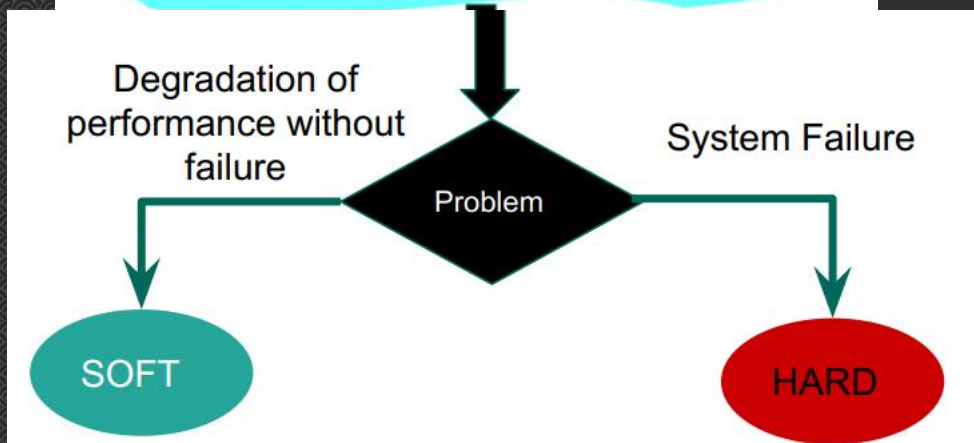
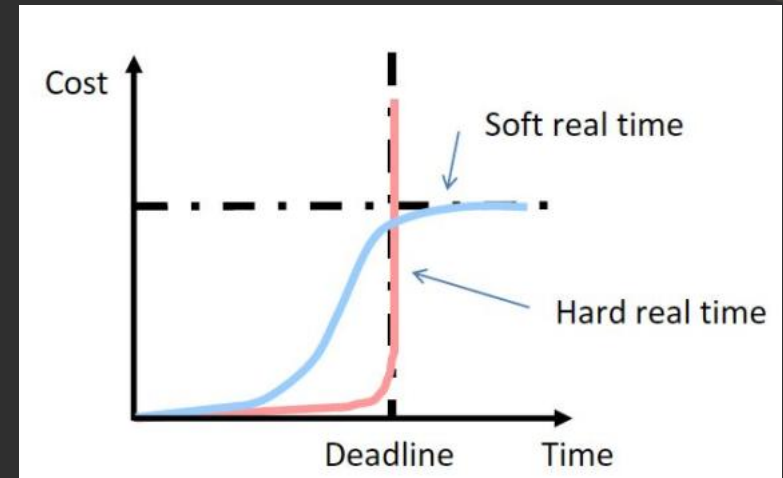
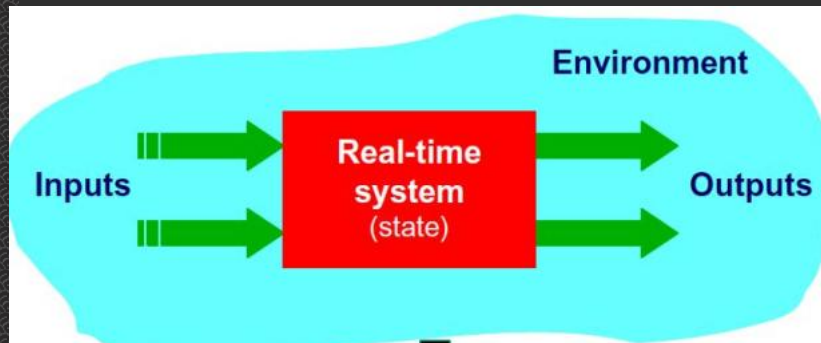
Appropriate execution time



- ◆ In real time embedded systems, certain task must be performed within a specified period of time.
- ◆ Normally desktop pc cannot achieve real time performance.
- ◆ In some cases, failure to meet a deadline is unsafe and can even endanger lives
- ◆ Therefore, special operating system known as real time operating systems run on these embedded systems.
 - ◆ In hard real time embedded system deadlines has to be strictly met but
 - ◆ whereas in soft real time embedded system the task may not be performed in a timely manner.

!The software developer needs to ascertain whether the embedded system is a hard real time or soft one and has to perform the performance analysis accordingly

- ❖ A real-time system reacts to inputs from the environment and produces output that affect environment



System type	
Traffic light control	Hard RT—critical
Automated teller machine	Soft RT—noncritical
Controller for radiation therapy machine	Hard RT—critical
Car simulator for driver training	Hard RT—noncritical
Highway car counter	Soft RT—noncritical
Missile control	Hard RT—noncritical
Video games	Hard RT—critical
Network chat	Soft RT—noncritical

QUALITY ATTRIBUTES OF EMBEDDED SYSTEMS



- ❖ Quality attributes are the non-functional requirements that need to be documented properly in any system design
- ❖ If the quality attributes are more concrete and measurable it will give a positive impact on the system development process and the end product
- ❖ The various quality attributes that needs to be addressed in any Embedded System development are broadly classified into two, namely
 - ❖ ‘Operational Quality Attributes’ and
 - ❖ ‘Non-Operational Quality Attributes’

Operational Quality Attributes

- ◆ The operational quality attributes represent the relevant quality attributes related to the Embedded System when it is in the operational mode or ‘online’ mode.
- ◆ The important quality attributes coming under this category are listed below:
 1. Response
 2. Throughput
 3. Reliability
 4. Maintainability
 5. Security
 6. Safety

Operational Quality Attributes



1. Response

- ◆ Response is a measure of quickness of the system.
- ◆ It gives you an idea about how fast your system is tracking the changes in input variables.
- ◆ Most of the embedded systems demand fast response which should be almost Real Time.
- ◆ For example, an embedded system deployed in flight control application
 - ◆ should respond in a Real Time manner.
 - ◆ Any response delay in the system will create potential impact to the safety of the flight as well as the passengers. It is not necessary that all embedded systems should be Real Time in response.
- ◆ For example, the response time requirement for an electronic toy
 - ◆ is not at all time-critical.
 - ◆ There is no specific deadline that this system should respond within this particular timeline

Operational Quality Attributes



2. Throughput

- ◆ Throughput deals with the efficiency of a system.
- ◆ In general it can be defined as the rate of production or operation of a defined process over a stated period of time.
- ◆ The rates can be expressed in terms of units of products, batches produced, or any other meaningful measurements.
- ◆ In the case of a Card Reader,
 - ◆ throughput means how many transactions the Reader can perform in a minute or in an hour or in a day.
- ◆ Throughput is generally measured in terms of 'Benchmark'.
 - ◆ A 'Benchmark' is a reference point by which something can be measured.
 - ◆ Benchmark can be a set of performance criteria that a product is expected to meet or a standard product that can be used for comparing other products of the same product line.

Operational Quality Attributes.....



3. Reliability

- ◆ Reliability is a measure of how much % you can rely upon the proper functioning of the system or what is the % susceptibility of the system to failures.
- ◆ Mean Time Between Failures (MTBF) and Mean Time To Repair (MTTR) are the terms used in defining system reliability.
- ◆ MTBF gives the frequency of failures in hours/weeks/months.
- ◆ MTTR specifies how long the system is allowed to be out of order following a failure.
- ◆ For an embedded system with critical application need, it should be of the order of minutes

Example 1: For example, let's say you want to calculate the MTBF of a X device that operates for 8 hours per day, 5 days a week, for a total of 1 year. During this time, the X device fails 4 times. To calculate the MTBF

Operational Quality Attributes.....

3. Reliability.....

◆ **Sol:**

- ◆ **Total operating time** = 8 hours/day x 5 days/week x 52 weeks = 2,080 hours
- ◆ **Number of failures** = 4
- ◆ **MTBF** = Total operating time / number of failures = 2,080 hours / 4 = 520 hours
- ◆ The MTBF of the X device is **520 hours**.
- ◆ This means that on average, the X device can be expected to operate for 520 hours before it fails.
- ◆ In reality, it might fail sooner, or later than 520 hours, and we won't understand *why* the motor is failing, but this average time is a useful metric

Operational Quality Attributes.....

4.

Maintainability

- ◆ Maintainability deals with support and maintenance to the end user or client in case of technical issues and product failures or on the basis of a routine system checkup.
- ◆ Reliability and maintainability are considered as two complementary disciplines.
- ◆ A more reliable system means a system with less corrective maintainability requirements and vice versa.
- ◆ As the reliability of the system increases, the chances of failure and non-functioning also reduces, thereby the need for maintainability is also reduced.
- ◆ Maintainability is closely related to the system availability
- ◆ Maintainability can be broadly classified into two categories, namely:
 - ◆ ‘Scheduled or Periodic Maintenance (preventive maintenance)’ and
 - ◆ ‘Maintenance to unexpected failures (corrective maintenance)’
- ◆ Some embedded products may use consumable component or may contain components which are subject to wear and tear and they should be replaced on a periodic basis.
- ◆ Example :
 - ◆ An inkjet printer uses ink cartridges, which are consumable components and as per the printer manufacturer the end user should replace the cartridge after each ‘n’ number of printouts to get quality prints
 - ◆ This is an example for ‘Scheduled or Periodic maintenance’

Operational Quality Attributes.....

5. Security

- ◆ ‘Confidentiality’, ‘Integrity’, and ‘Availability’ (The term ‘Availability’ mentioned here is not related to the term ‘Availability’ mentioned under the ‘Maintainability’ section) are the three major measures of information security.
- ◆ Confidentiality deals with the protection of data and application from unauthorized disclosure
- ◆ Integrity deals with the protection of data and application from unauthorized modification
- ◆ Availability deals with protection of data and application from unauthorised users
- ◆ Example:
- ◆ A very good example of the ‘Security’ aspect in an embedded product is a Personal Digital Assistant (PDA).
- ◆ The PDA can be either a shared resource (e.g. PDAs used in LAB setups) or an individual one.
 - ◆ If it is a shared one there should be some mechanism in the form of a user name and password to access into a particular person’s profile—This is an example of ‘Availability’
 - ◆ Also all data and applications present in the PDA need not be accessible to all users
 - ◆ Some of them are specifically accessible to administrators only

Operational Quality Attributes.....



5. Safety

- ❖ ‘Safety’ and ‘Security’ are two confusing terms.
- ❖ Sometimes you may feel both of them as a single attribute.
- ❖ But they represent two unique aspects in quality attributes.
- ❖ Safety deals with the possible damages that can happen to the operators, public and the environment due to the breakdown of an embedded system or due to the emission of radioactive or hazardous materials from the embedded products
- ❖ The breakdown of an embedded system may occur due to a hardware failure or a firmware failure
- ❖ Safety analysis is a must in product engineering to evaluate the anticipated damages and determine the best course of action to bring down the consequences of the damages to an acceptable level

Note That some of the safety threats are sudden (like product breakdown) and some of them are gradual (like hazardous emissions from the product).

Non-Operational Quality Attributes

- ❖ The quality attributes that needs to be addressed for the product 'not' on the basis of operational aspects are grouped under this category.
- ❖ The important quality attributes coming under this category are listed below.
 1. Testability & Debug-ability
 2. Evolvability
 3. Portability
 4. Time to prototype and market
 5. Per unit and total cost

Non-Operational Quality Attributes.....



1. Testability & Debug-ability

- ◆ ‘Testability deals with how easily one can test the design, application and by which means he/she can test it.
- ◆ For an embedded product, testability is applicable to both the embedded hardware and firmware
- ◆ Embedded hardware testing ensures that the peripherals and the total hardware functions in the desired manner, whereas firmware testing ensures that the firmware is functioning in the expected way
- ◆ Debug-ability is a means of debugging the product as such for figuring out the probable sources that create unexpected behaviour in the total system
- ◆ Debug-ability has two aspects in the embedded system development context, namely, hardware level debugging and firmware level debugging
- ◆ Hardware debugging is used for figuring out the issues created by hardware problems whereas firmware debugging is employed to figure out the probable errors that appear as a result of flaws in the firmware

Non-Operational Quality Attributes.....



2. Evolvability

- ◆ Evolvability is a term which is closely related to Biology.
- ◆ Evolvability is referred as the non-heritable variation.
- ◆ For an embedded system, the quality attribute 'Evolvability' refers to the ease with which the embedded product (including firmware and hardware) can be modified to take advantage of new firmware or hardware technologies

3. Portability

- ◆ Portability is a measure of 'system independence'
- ◆ An embedded product is said to be portable if the product is capable of functioning 'as such' in various environments, target processors/ controllers and embedded operating systems.
- ◆ The ease with which an embedded product can be ported on to a new platform is a direct measure of the re-work required
- ◆ A standard embedded product should always be flexible and portable.
- ◆ In embedded products, the term 'porting' represents the migration of the embedded firmware written for one target processor (e.g. Intel x86) to a different target processor (say an ARM Cortex M3 processor from Freescale

Non-Operational Quality Attributes.....



4. Time-to-Prototype and Market

- ◆ Time-to-market is the time elapsed between the conceptualisation of a product and the time at which the product is ready for selling (for commercial product) or use (for non-commercial products)
- ◆ The commercial embedded product market is highly competitive and time to market the product is a critical factor in the success of a commercial embedded product.
- ◆ There may be multiple players in the embedded industry who develop products of the same category (like mobile phone, portable media players, etc.)
- ◆ If you come up with a new design and if it takes long time to develop and market it, the competitor product may take advantage of it with their product.
- ◆ Embedded technology is one where rapid technology change is happening
- ◆ Product prototyping helps a lot in reducing time-to-market

Non-Operational Quality Attributes.....



5. Per Unit Cost and Revenue

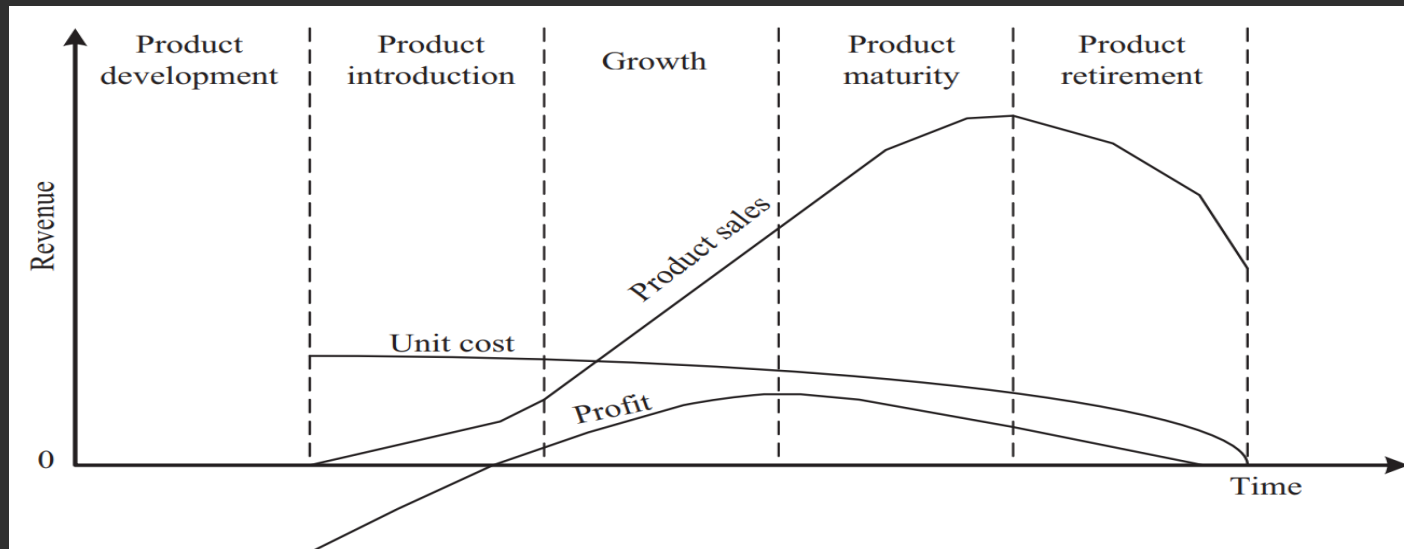
- ◆ Cost is a factor which is closely monitored by both end user (those who buy the product) and product manufacturer (those who build the product)
- ◆ Cost is a highly sensitive factor for commercial products
- ◆ Any failure to position the cost of a commercial product at a nominal rate may lead to the failure of the product in the market
- ◆ Proper market study and cost benefit analysis should be carried out before taking a decision on the per-unit cost of the embedded product
- ◆ From a designer/product development company perspective the ultimate aim of a product is to generate marginal profit
- ◆ The budget and total system cost should be properly balanced to provide a marginal profit
- ◆ The Product Life Cycle (PLC)

Non-Operational Quality Attributes.....



- ◆ The Product Life Cycle (PLC)
 - ◆ Embedded system product life cycle
 - ◆ Starts with the design and development phase
 - ◆ And full activity stage to be carried out are
 - ◆ Product idea generation,
 - ◆ prototyping,
 - ◆ Roadmap definition,
 - ◆ actual product design and development
 - ◆ are the activities carried out during this phase
 - ◆ During the design and development phase there is only investment and no returns
 - ◆ At this stage is only investment and no returns
 - ◆ Product Introduction stage
 - ◆ At this stage the product is ready to sell, it is introduced to the market
 - ◆ During this initial period the sales and revenue will be low
 - ◆ There won't be much competition and the product sales and revenue increases with time
 - ◆ Growth
 - ◆ The growth phase, the product grabs high market share

- ◆ The Product Life Cycle (PLC)....
 - ◆ Maturity phase
 - ◆ the growth and sales will be steady and the revenue reaches at its peak
 - ◆ The Product Retirement
 - ◆ it's a decline phase which starts with the drop in sales volume, market share and revenue
 - ◆ The decline happens due to various reasons like competition from similar product with enhanced features or technology changes , etc
 - ◆ At some point of the decline stage, the manufacturer announces discontinuing of the product
- ◆ The different stages of the embedded products life cycle—revenue, unit cost and profit in each stage—are illustrated in the following Product Life-cycle graph



Product life cycle (PLC) curve

Non-Operational Quality Attributes.....



The Product Life Cycle (PLC)...

- ◆ From the graph, it is clear that the total revenue increases from the product introduction stage to the product maturity stage
- ◆ The revenue peaks at the maturity stage and starts falling in the decline/retirement stage
- ◆ The unit cost is very high during the introductory stage
- ◆ Typical example is cell phone;
 - ◆ if you buy a new model of cell phone during its launch time, the price will be high and you will get the same model with a very reduced price after three or four months of its launching

Non-Operational Quality Attributes.....



- ◆ Non-Recurring Engineering(NRE) costs
 - ◆ NRE is that cost associated with developing a product.
 - ◆ are the base of most technology managers'lives.
 - ◆ Its converse is the cost of Goods Sold(COGS), a.k.a Recurring Costs
 - ◆ NRE costs are amortized over the life of a product in fact or in reality
 - ◆ Example
 - ◆ Mature companies carefully compute the amount of engineering a the product—a car maker, for instance, might spend a billion bucks engineering a new model with a lifespan of a million units sold; in this case the cost of the car goes up by \$1000 to pay for the NRE.
 - ◆ Increasing NRE costs drives up the product ' s price (most likely making it less competitive and thus reducing profits), or directly reduces profits

Further reading



- ◆ <https://www.mdpi.com/2076-3417/13/7/4608>
- ◆ <https://www.bmc.com/blogs/mtbf-vs-mtff-vs-mttr-whats-difference/>
- ◆ <https://inxee.com/blog/5-functional-attributes-in-architecture-of-any-embedded-system/>
- ◆ <https://intechhouse.com/blog/quality-attributes-in-embedded-systems-how-to-build-reliable-and-resilient-devices>
- ◆ cselectricalandelectronics.com/top-15-applications-of-embedded-systems-embedded-system-application/
- ◆ <https://www.theengineeringprojects.com/2021/06/characteristics-of-embedded-systems.html>
- ◆ [Embedded Product Development Life Cycle: Four Main Steps | SaM Solutions](#)
- ◆ [Embedded Development Life Cycle](#)
- ◆ <https://www.wellpcb.com/blog/guides/nre-cost/>
- ◆ <https://titoma.com/nre/>
- ◆ [Embedded System Design - an overview | ScienceDirect Topics](#)
- ◆ [Fast and Effective Embedded Systems Design | ScienceDirect](#)